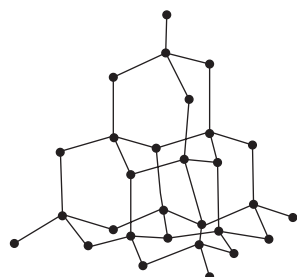
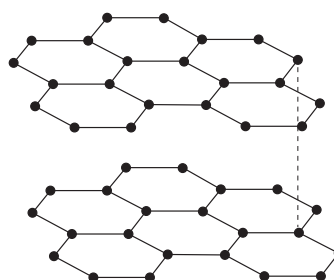


11. The diagrams show the structure of diamond and graphite.



diamond



graphite

(a) Name the atom being represented by a • in both diagrams.

[1]

.....

(b) Name the type of **bonding** found in both diamond and graphite.

[1]

.....



Examiner
only

(c) The table shows some properties of graphite.

Properties of graphite
soft
high melting point
insoluble in water
conducts electricity

Use only properties from the table to answer this question.

Give **two** properties of graphite that are **different** from those of diamond. Give a use relating to each property. [4]

Property 1

Use

Property 2

Use

END OF PAPER

6



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		



3 4 5 6 7 0

Key

atomic number